

The Optical Internet

Wavelength Division Multiplexing (WDM)

and Lambda Switching

Alessio Sacco, Marco Rapelli

Politecnico di Torino

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Introduction



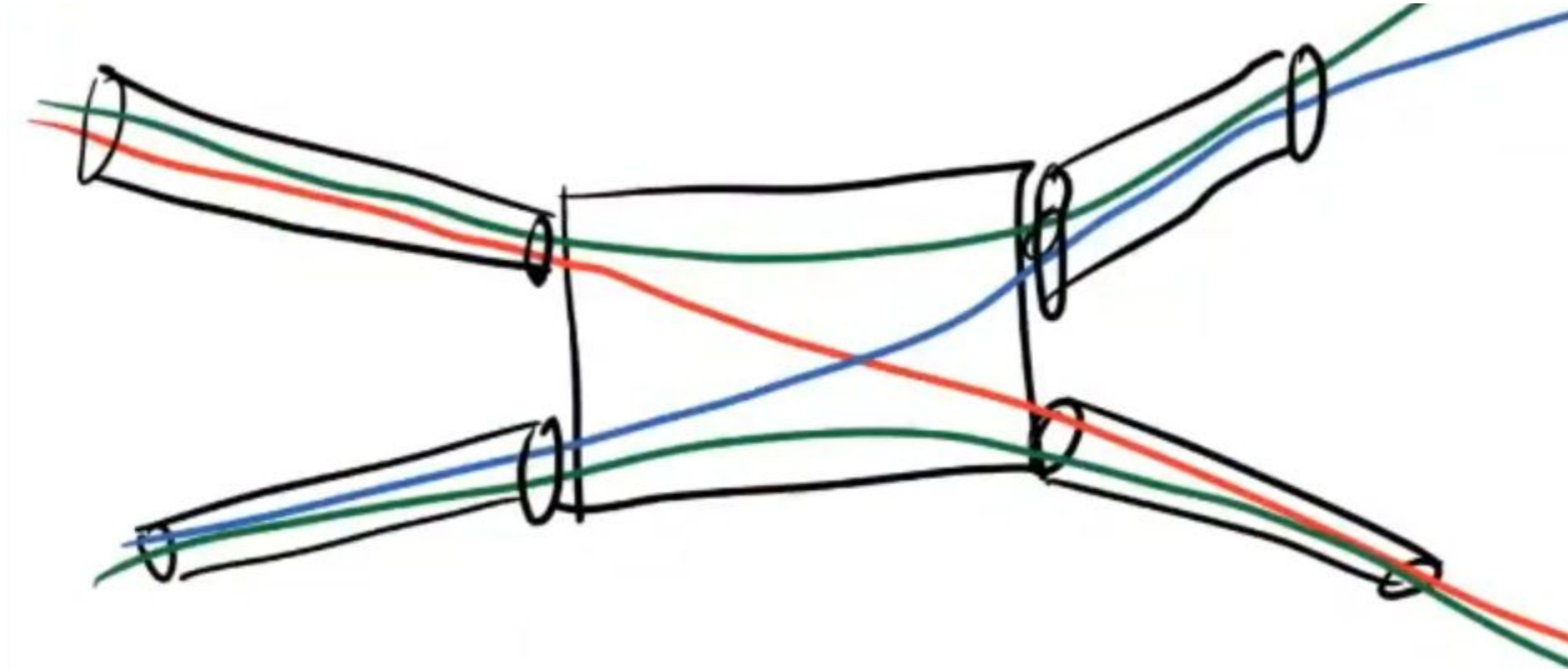
WDM: Wavelength Division Multiplexing

- Transmission of multiple light signals (wavelengths) on the same strand of fiber
- DWDM - Dense WDM
 - More sophisticated → more expensive
- CWDM - Coarse WDM
 - Lower number of wavelengths → cheaper

Initial WDM Application

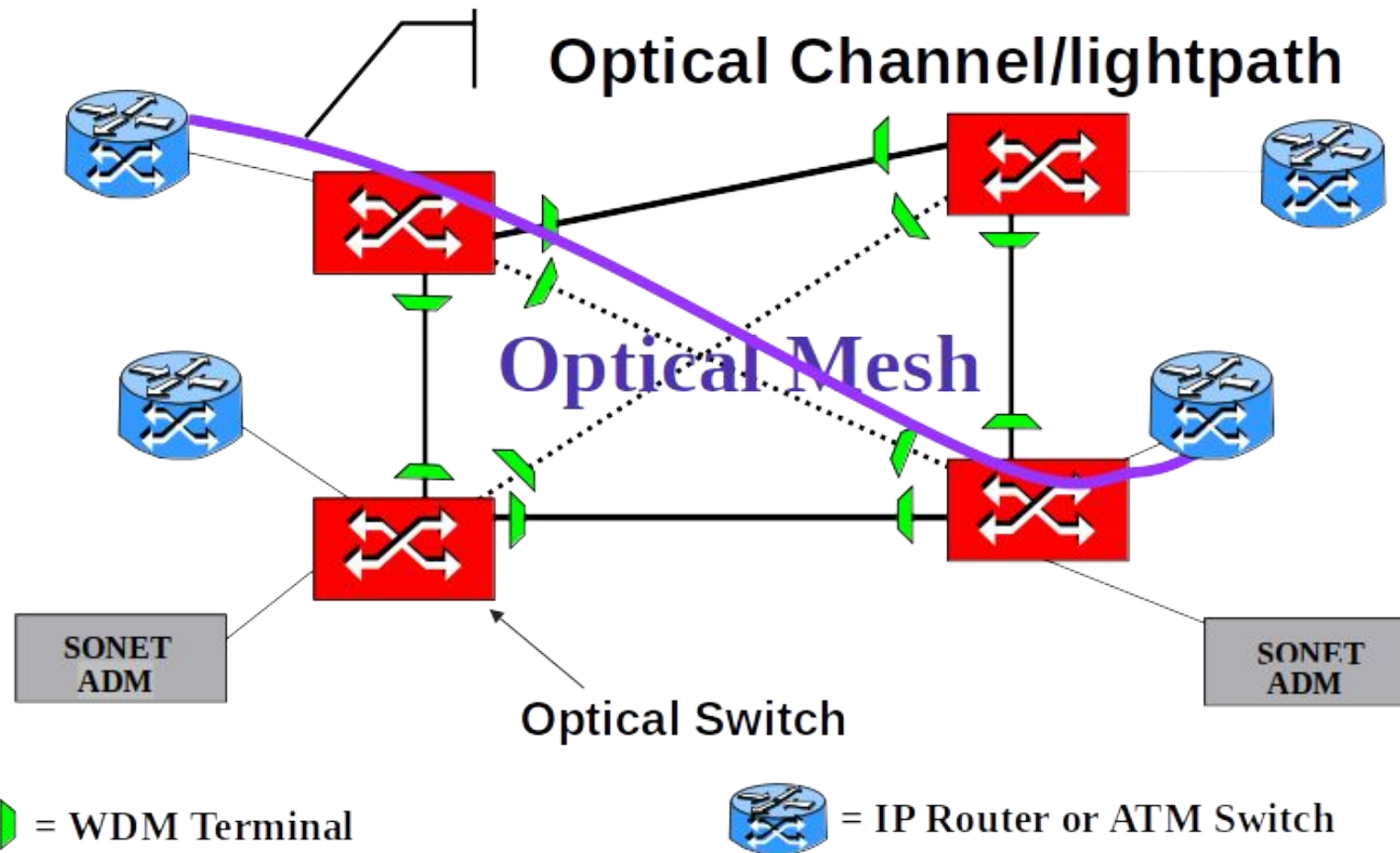
- Increase transmission capacity of fiber
- Trunk bandwidth
- Increase the utilization (ROI: return of investment) of [existing] fiber
- Point to point configurations

Wavelength Switch or Lambda Switch



What to do with Optical Switches?

- Deployed in the network core because of its coarse bandwidth allocation
- 1 optical channel: 2.4Gb/s or more



Why Optical Switching?

- It has the potential of being simple, hence delivering a (very) low cost per switched bit

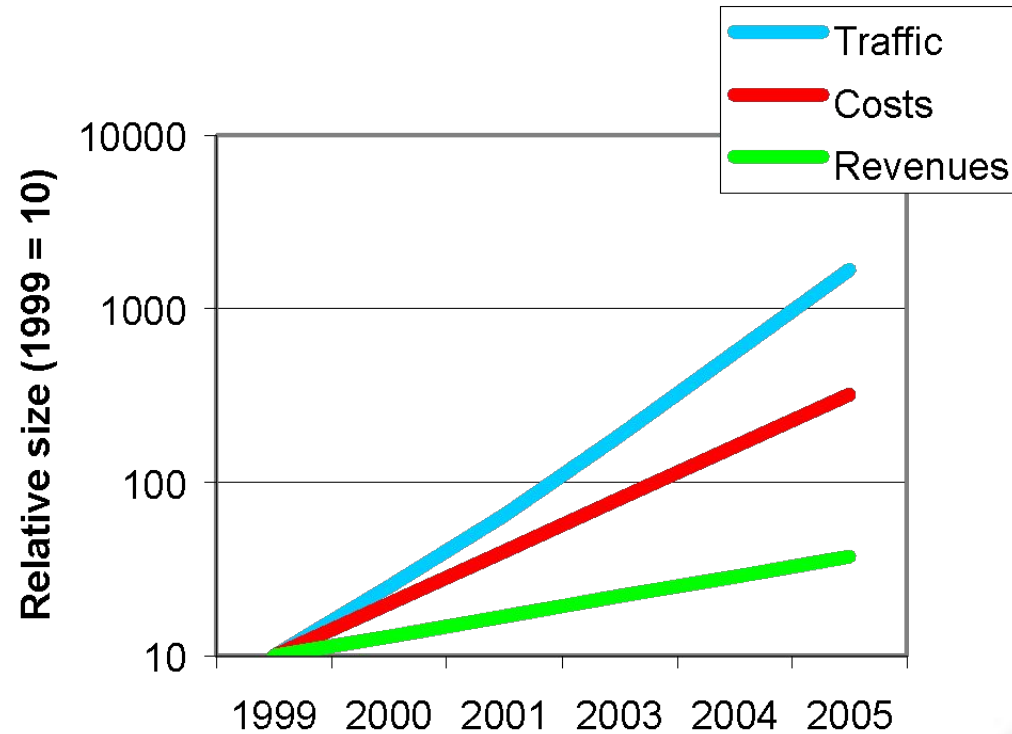
A bit of context



The Resulting Traffic Dilemma...

**Traffic is
growing at
explosive rates**

**...And network
costs are
growing faster
than revenues**

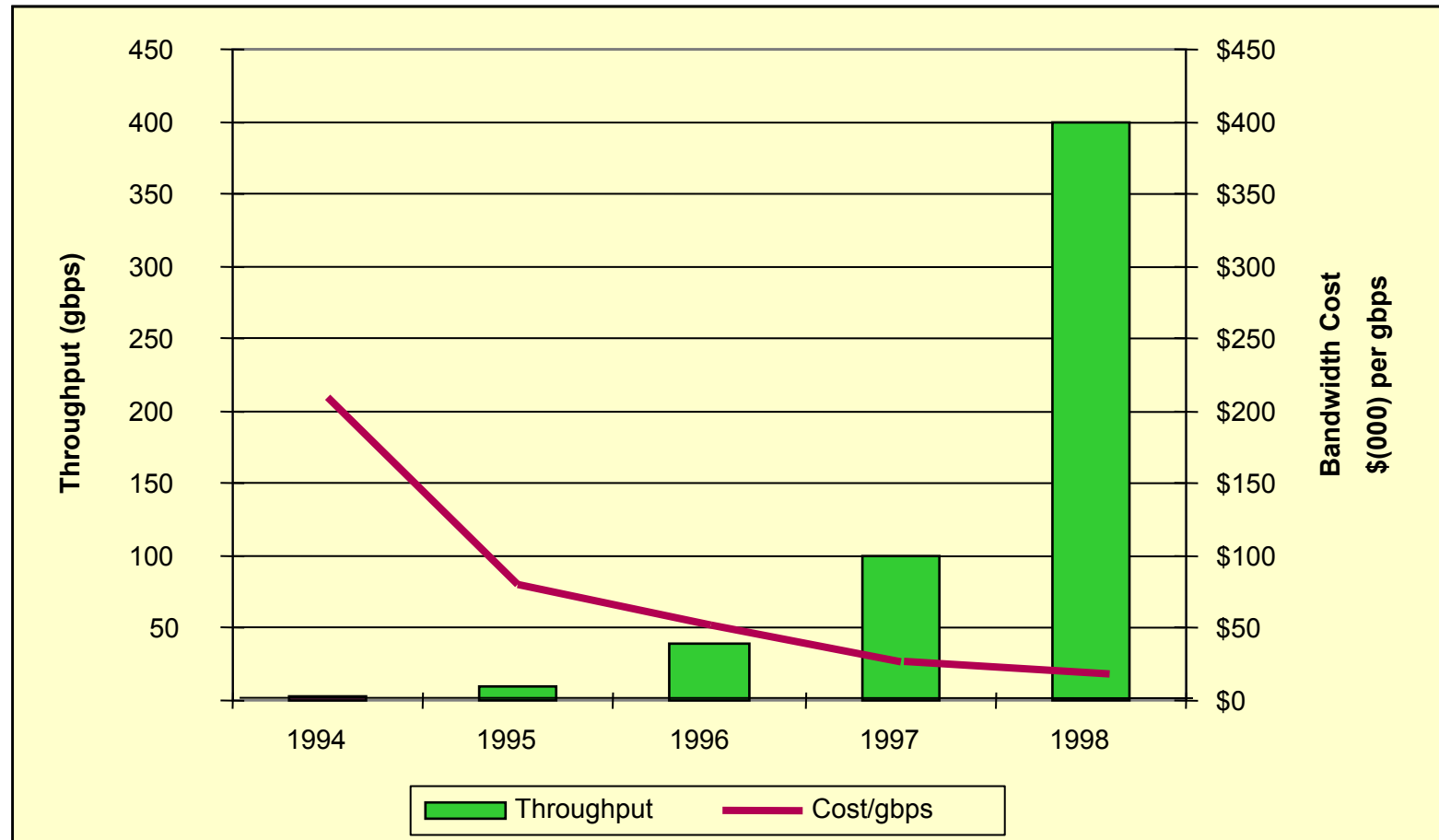


- Optical networks help solve this dilemma -- on a large scale.

RHK

Optics Reshape the Cost Curve

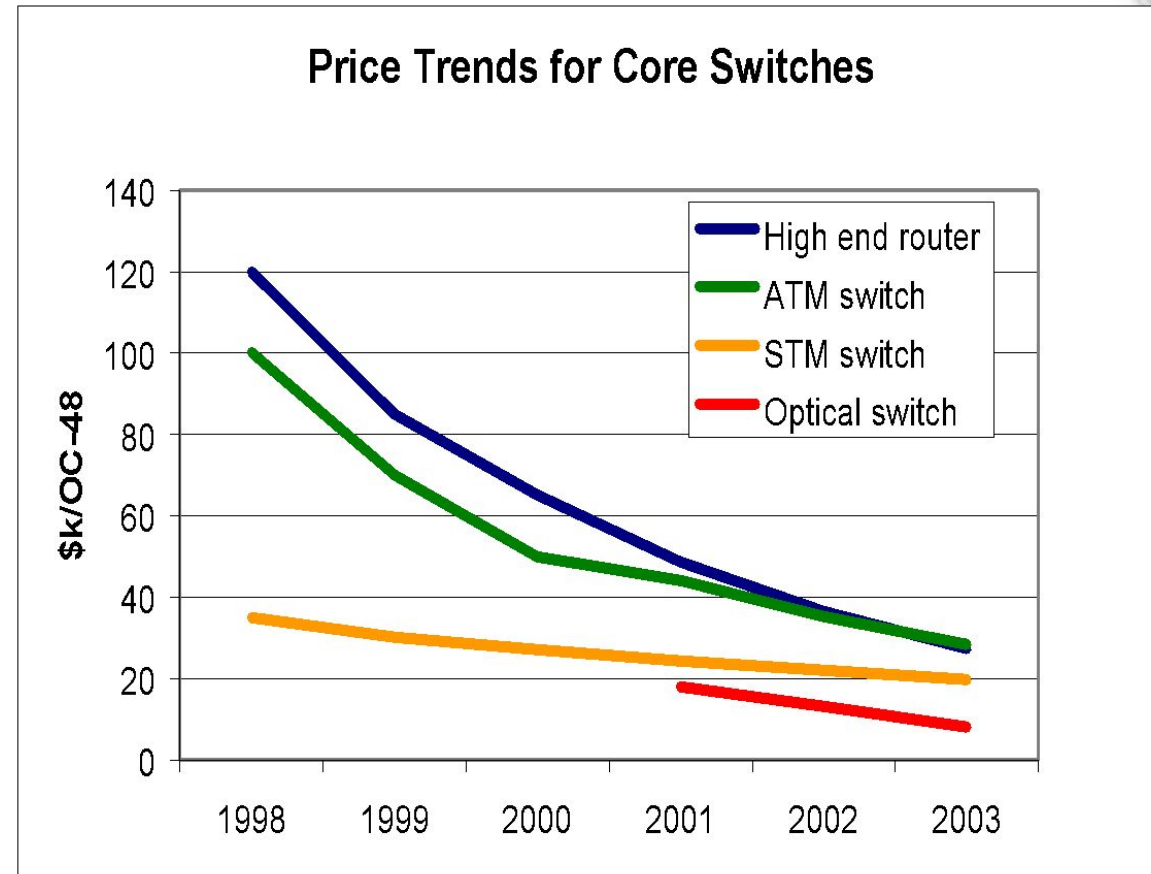
- DWDM driving down the cost of trunk bandwidth



RHK

Pricing & The Technologies

- **Prices/OC-48 (2.5Gbps) dropping rapidly – in favor of 10Gbps+ systems**
- **By 2002, optical switches could offer the most bandwidth per dollar**



Back to the future

(and to technology, leaving hype behind)



Many Flavors of Optical Switches

- Optical vs Electronic core
- Cross Connect vs Switch
- Wavelength Conversion
- Different levels of complexity
- Different levels of flexibility



Switching Core



Optical Core

- Deploy physical properties of materials to deflect light from incoming fiber to outgoing fiber
- Tilting mirrors
 - Micro-electro-mechanical systems (MEMS)
 - Voltage operated
- Holographic reflecting surfaces
 - Voltage operated
- Materials changing properties with
 - Heat
 - Pressure
 - Voltage/current



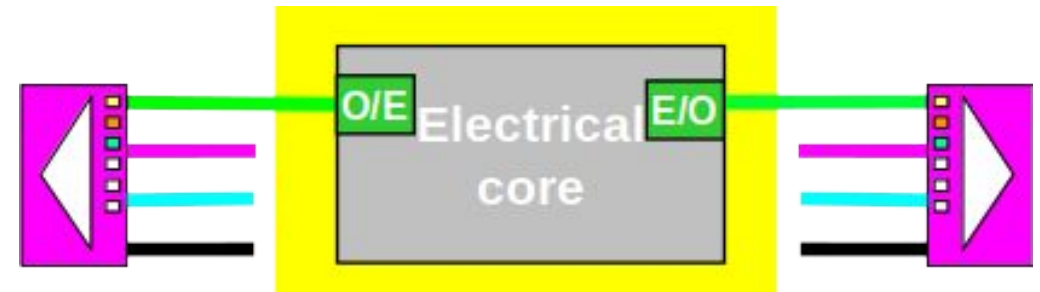
Optical Core Properties

- [Potentially] inexpensive (low CAPEX)
 - Low cost material
 - Low cost process [once technology is mature]
- Bit rate and signal independent
 - Unlimited scalability
 - Multi standard
- Low power consumption
 - Low cost of operation (OPEX)
- High production costs
 - Immature technology
- High attenuation (and no regeneration)



Electrical Core

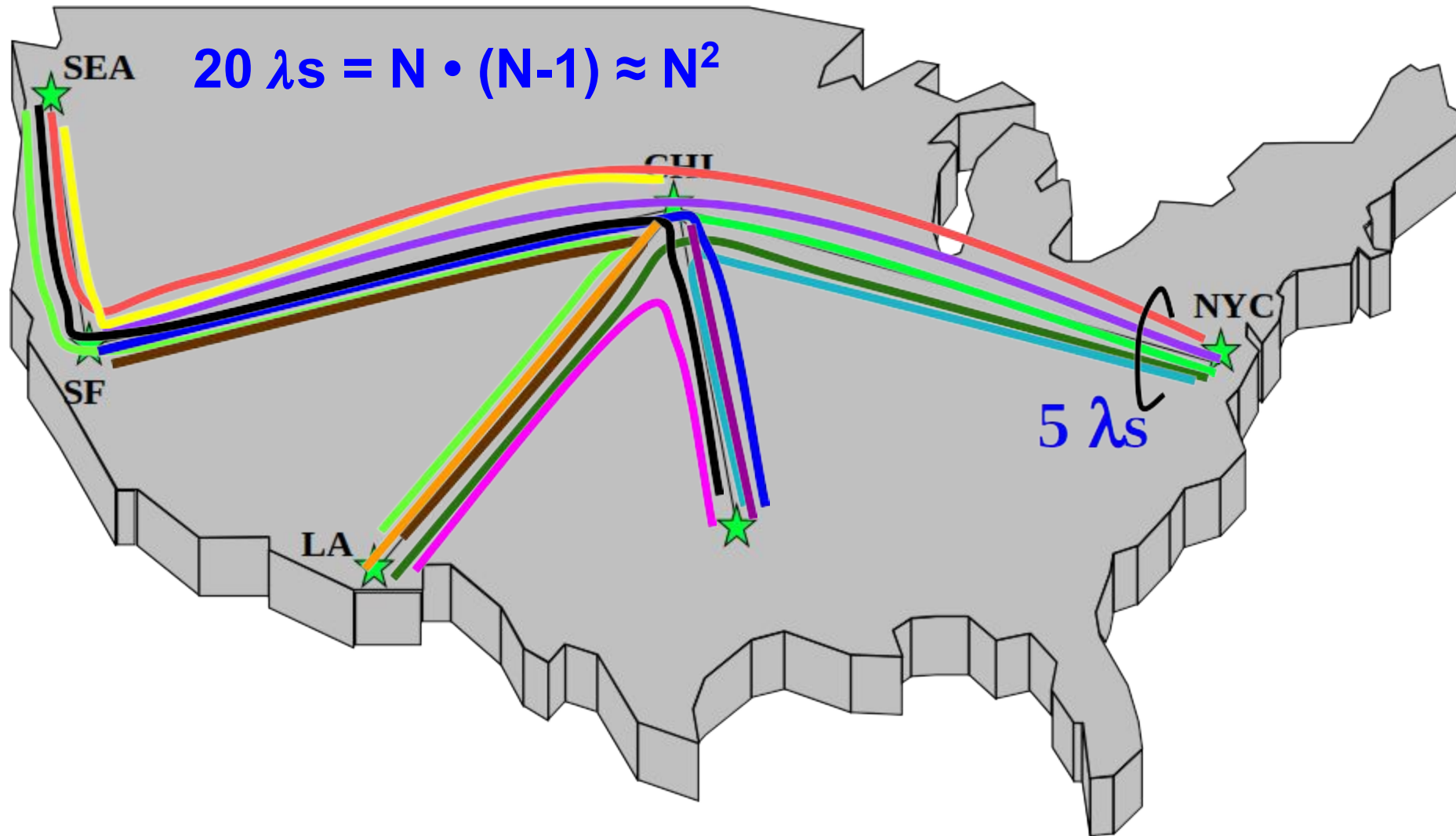
- Convert optical signal into an electric one and use a circuit interconnection network
- Optical-electrical conversion
- Receive the bits and switch them
- It loses all the nice properties of an optical core
 - Bit rate independence
 - Low power consumption
 - Low cost
- However
 - At current state of technology, cheaper
 - Less complex/costly than packet switching



Wavelength Conversion



λ Switching: the N^2 Problem



Wavelength Conversion

- Complex
 - OEO conversion
 - Expensive
 - Non data transparent → does not scale
 - Physical properties
 - E.g., resonance chamber
 - Immature technology → expensive
- Does not require the same wavelength end-to-end
- No wavelength assignment problem
 - N^2 problem

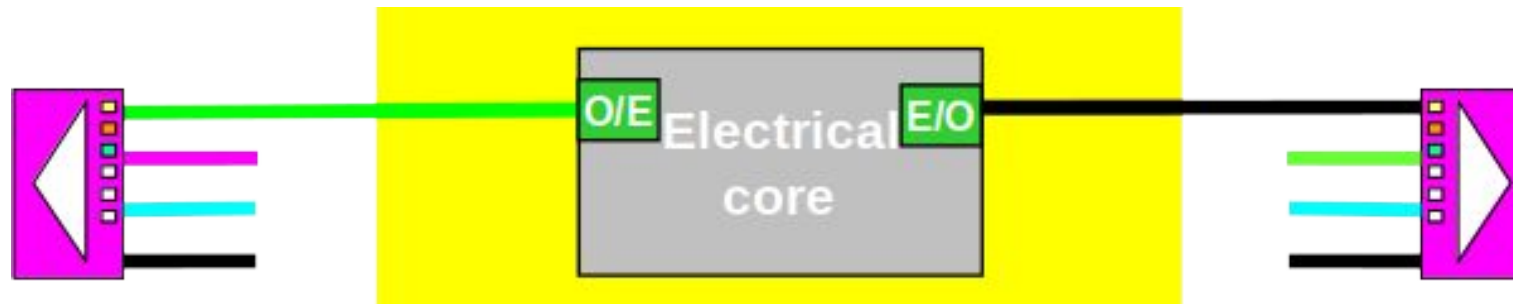


Common Combinations



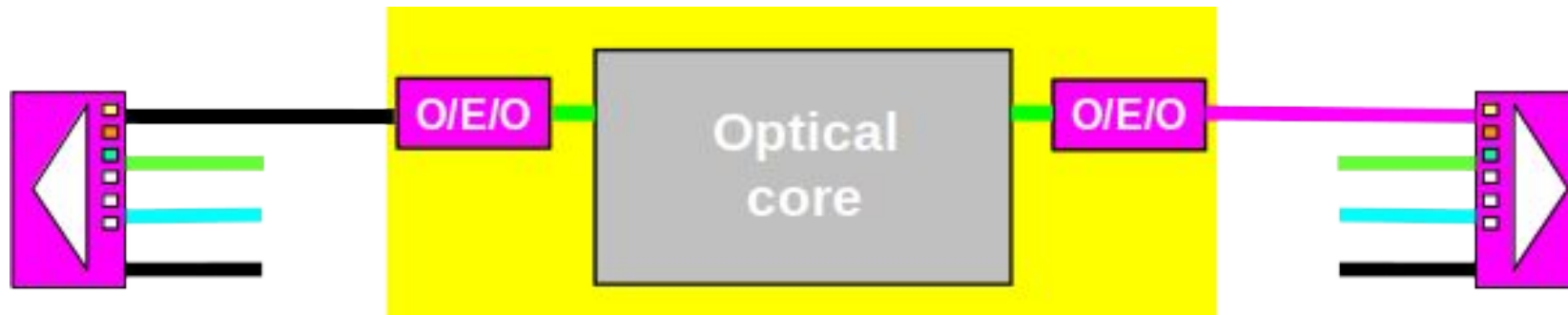
Wavelength Conversion

- One (or more) wavelengths from an input fiber to other one (or others) on an output fiber
- Electrical core might be used
 - Easier signal monitoring
 - Forward error correction (FEC) possible to reduce Bit Error Ratio (BER)



Wavelength Conversion

- Optical core with OEO (optical-electrical-optical) conversion
 - Also providing signal regeneration



Dynamic Optical Switching

- Wavelength switch with or without wavelength conversion
- Switch configuration is changed dynamically
 - By management
 - By time of day
 - By end system signaling
 - Every packet!?! ...
 - Optical packet switching
 - Optical burst switching



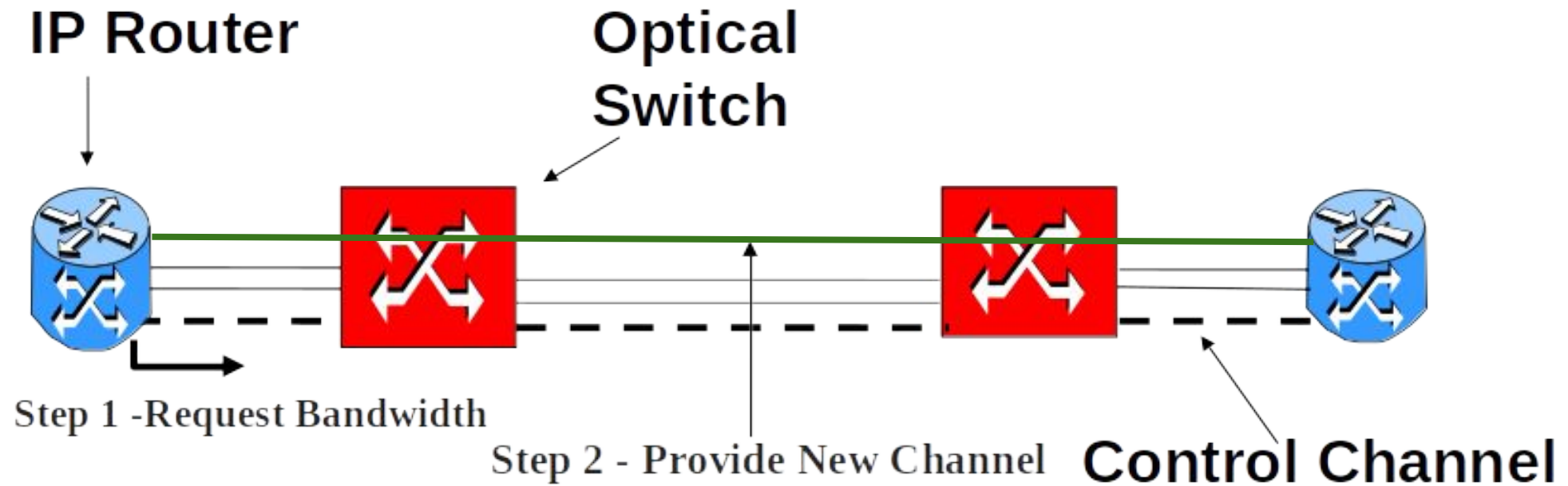
Deployment



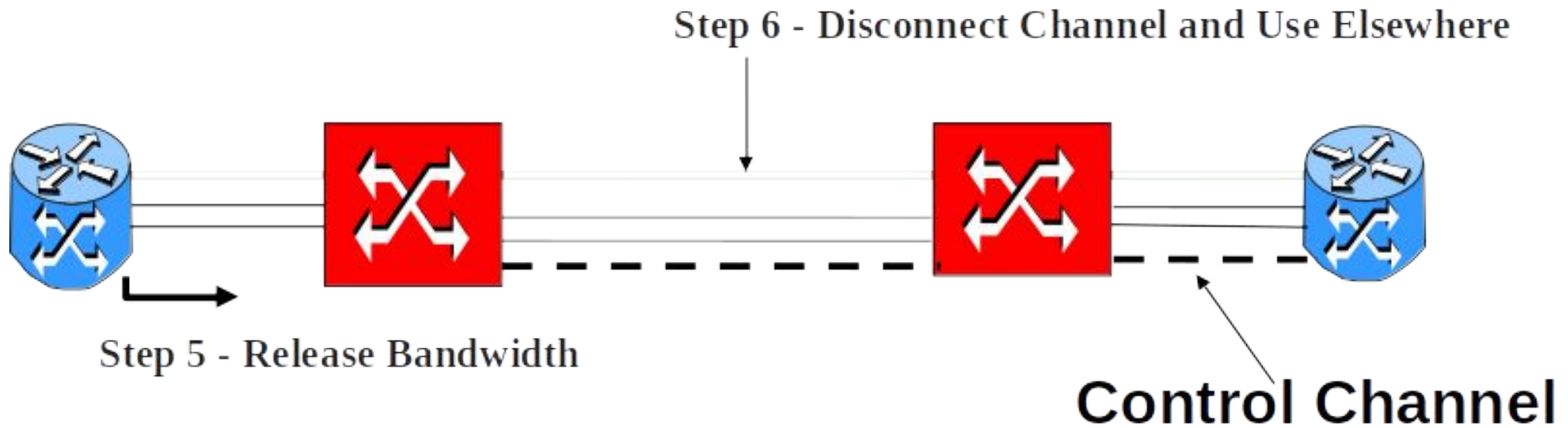
What is Expected from the Optical Network?

- Provisioning and protection of lightpaths end-to-end
- Client equipment (e.g. routers) to control provisioning of optical layer lightpaths
 - Signaling
- Cost-effective deployment of flexible networks

Provisioning

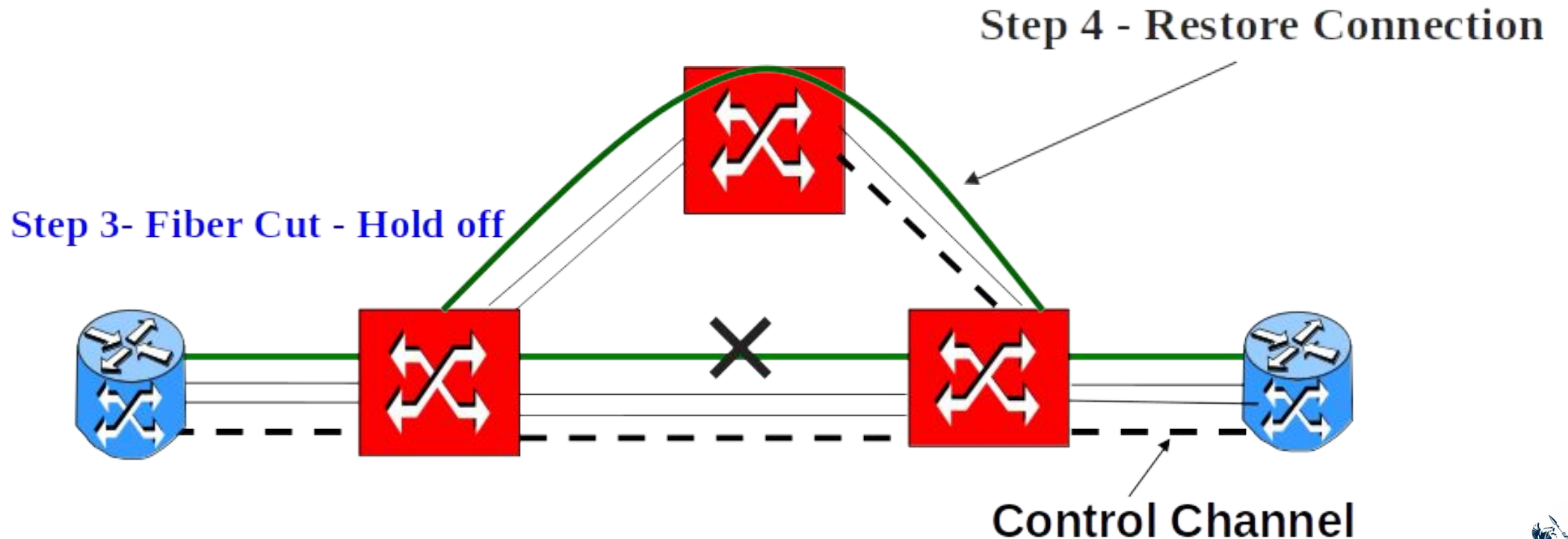


Provisioning



Protection/Restoration

- Protection: pre-determined action
 - non-optimal resource utilization
- Restoration: dynamically determined action
 - optimization of resource utilization



Protection/Restoration

- Multiple levels of protection:
 - Layer 1 optical, e.g. SONET-like
 - Layer 2 data link bundle
 - Layer 2.5 protected MPLS LSPs
 - Layer 3 routing
- Multiple layers of restoration can be triggered
 - Each different timescales for detection and repair
- Must avoid:
 - Unnecessary traffic shifting
 - Packet loss, reordering, control plane churn
 - Pathological feedback
 - Non self-stabilizing

Control Plane



What Optical Switches Need

- Resource discovery
 - Topology
 - Access points and node identification
 - Resource usage
- Connection management/signaling
 - Lightpath setup
 - Lightpath take down
 - Lightpath modification
- Distributed routing
- Mesh/ring network protection and recovery
- Establishment of protection service classes



What Optical Network Users Need

- Resource discovery
 - Address of users reachable through the optical network
- Manage lightpaths
 - Lightpath setup
 - Lightpath take down
 - Lightpath modification
- Negotiate protection service classes
 - Protected, unprotected, best effort lightpaths
- Does all this sound familiar? **ATM**



Routing

- In the Optical Internet network users are routers
- Overlay Model
 - The optical network provides connectivity between routers
 - Routers see the optical network as a black box
 - Routers might be provided with reachability information
- Peer Model
 - Routers and switches participate to the same routing protocols
 - Routers know the topology of the optical network
 - Routers can choose the preferred path for lightpaths between them
 - To reach specific destinations



How to Do It

- How is the optical network controlled?
 - Layer 3 control plane?
 - MPLS/LDP?
 - LSPs mapped over wavelengths
 - OSPF, BGP4?
 - New signaling and routing standards?
 - Proprietary vendor specific?
- Out of band or in-band
 - Ethernet control channel



MPLS choice for the Optical Network Control Plane

- MP λ S - Multi-Protocol Lambda Switching
- OSPF, IS-IS, BGP for resource discovery
- RSVP/LDP for signaling

Data Transport and Protocol Stack



Some Encapsulation Options

